Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 9

Lecture 17

17.1 Grover's algorithm with Unknown number of solutions

17.2 Quantum counting

17.3 Quantum counting example

Lecture 18 Quadratic Unconstrained Binary Optimization (QUBO)

18.1 Optimization Problems

18.2 Number Partitioning

18.3 Graph Partitioning

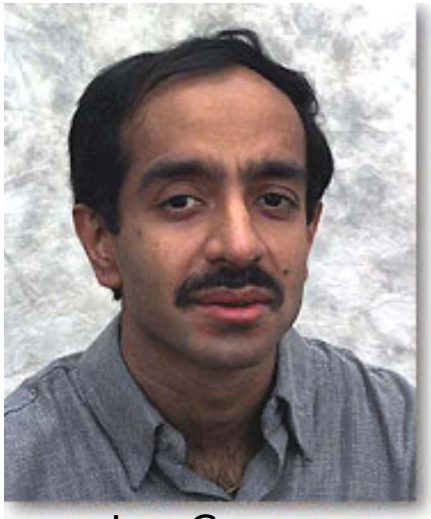
18.4 Traveling Salesman Problem

Practice class 9

Quantum counting and QUBO

Recap: Grover's Algorithm

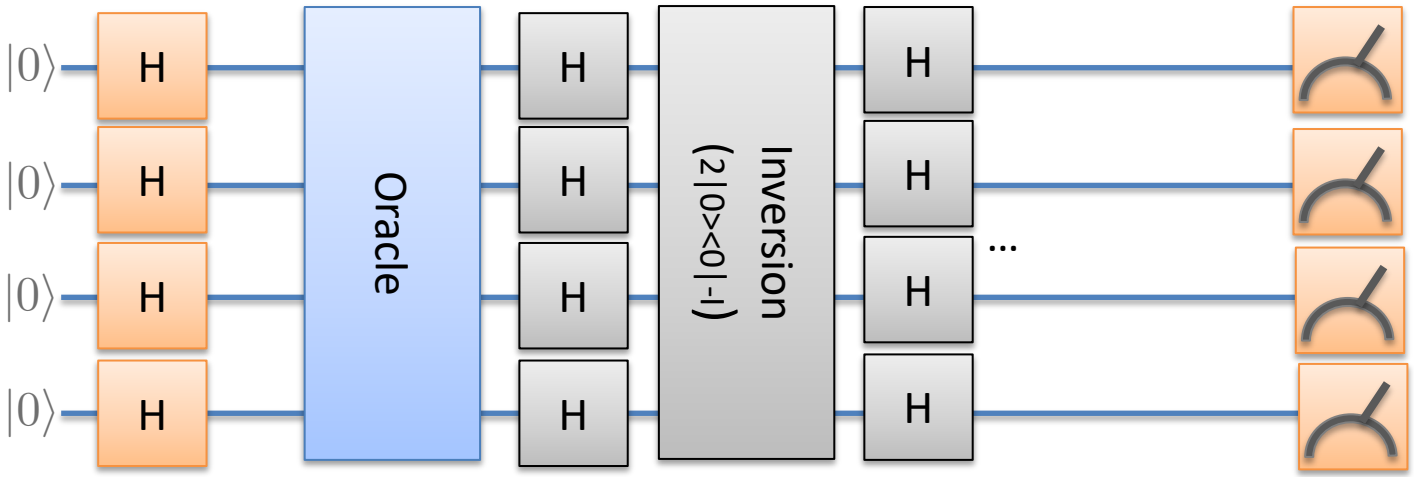
- Unordered search, find one marked item among many
- Classically, this requires N/2 queries to the oracle
- Quantum mechanically, requires only O(√N) queries.



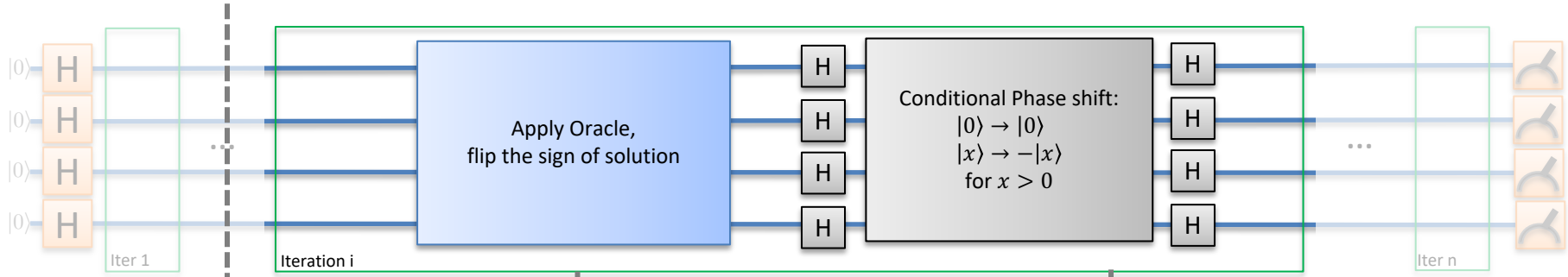
Lov Grover

Simple problem = search for one integer marked by the oracle.

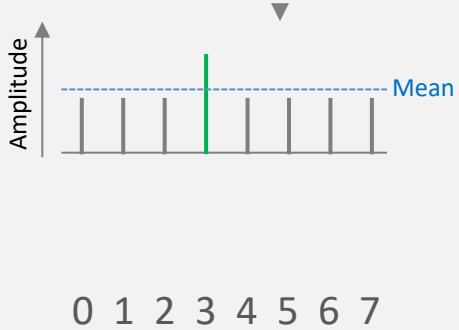
High level structure:

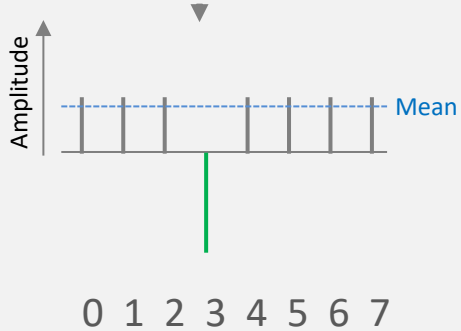


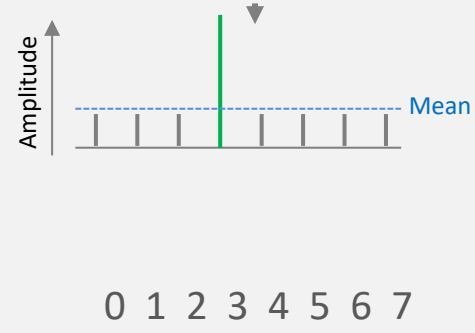
Recap: Grover's algorithm



Maps every amplitude Mean$-\delta$ to Mean$+\delta$





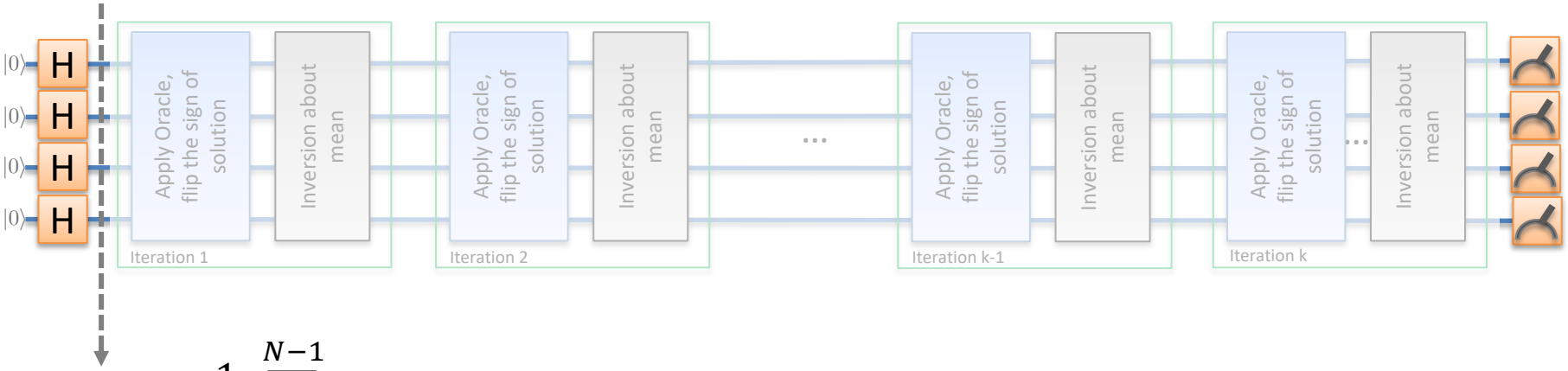


Intuition:

Flipping the sign of the amplitude of solution vector reduces the mean.

Inversion about the mean increases the amplitude of solution vector.

Recap: Number of Grover iterations



$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$ Probability $t = \frac{M}{N} = {m_0}^2$, where M is the number of solutions.

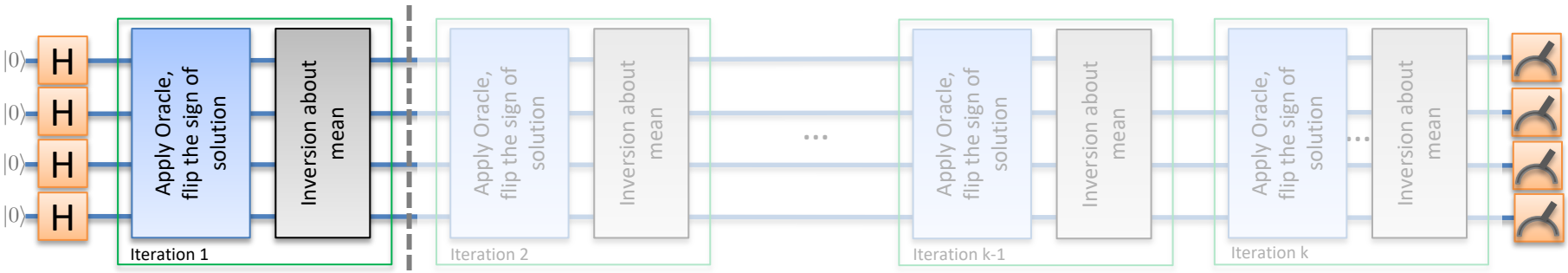
- Let $|\psi_M\rangle = \frac{1}{\sqrt{M}} \sum_{x \in S} |x\rangle$ and $|\psi_P\rangle = \frac{1}{\sqrt{P}} \sum_{x \notin S} |x\rangle$ where
  - S is the set of solutions, M is the number of solutions, and $P = N - M$.
  - $|\psi_M\rangle$ are the states that represent solution and $|\psi_P\rangle$ represent non-solution states.

$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle = \sqrt{\frac{M}{N}}\, |\psi_M\rangle + \sqrt{\frac{P}{N}}\, |\psi_P\rangle = m_0 |\psi_M\rangle + \mathrm{p}_0 |\psi_P\rangle$$

Each iteration changes the amplitudes.

Two basis states to understand Grover's algorithm!

Recap: Number of Grover iterations



Probability $t = m^2{}_{j+1} = sin^2\big((2j+1)\theta\big)$,

where j is the number of iterations and $\sin(\theta) = \sqrt{M/_N}$

- Inversion about the mean transforms:
  - j^{th} iteration: $-m_j|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle \rightarrow (2A_j\sqrt{M} + m_j)|\psi_M\rangle + (2A_j\sqrt{N-M} - p_j)|\psi_P\rangle$
    $= m_{j+1}|\psi_M\rangle + \mathrm{p}_{\mathrm{j+1}}|\psi_P\rangle$

- Solution to the recurrence equations:
  - $m_{j+1} = \sin\big((2j+1)\theta\big)$
  - $p_{j+1} = \cos\big((2j+1)\theta\big)$
  - $\sin(\theta) = \sqrt{\frac{M}{N}}$

How many iterations required?

$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

For small angles,

$$\theta \approx \frac{\sqrt{M}}{\sqrt{N}}$$

After k iterations, we rotate to have only marked solutions:

$$(2k+1)\theta = \frac{\pi}{2}$$

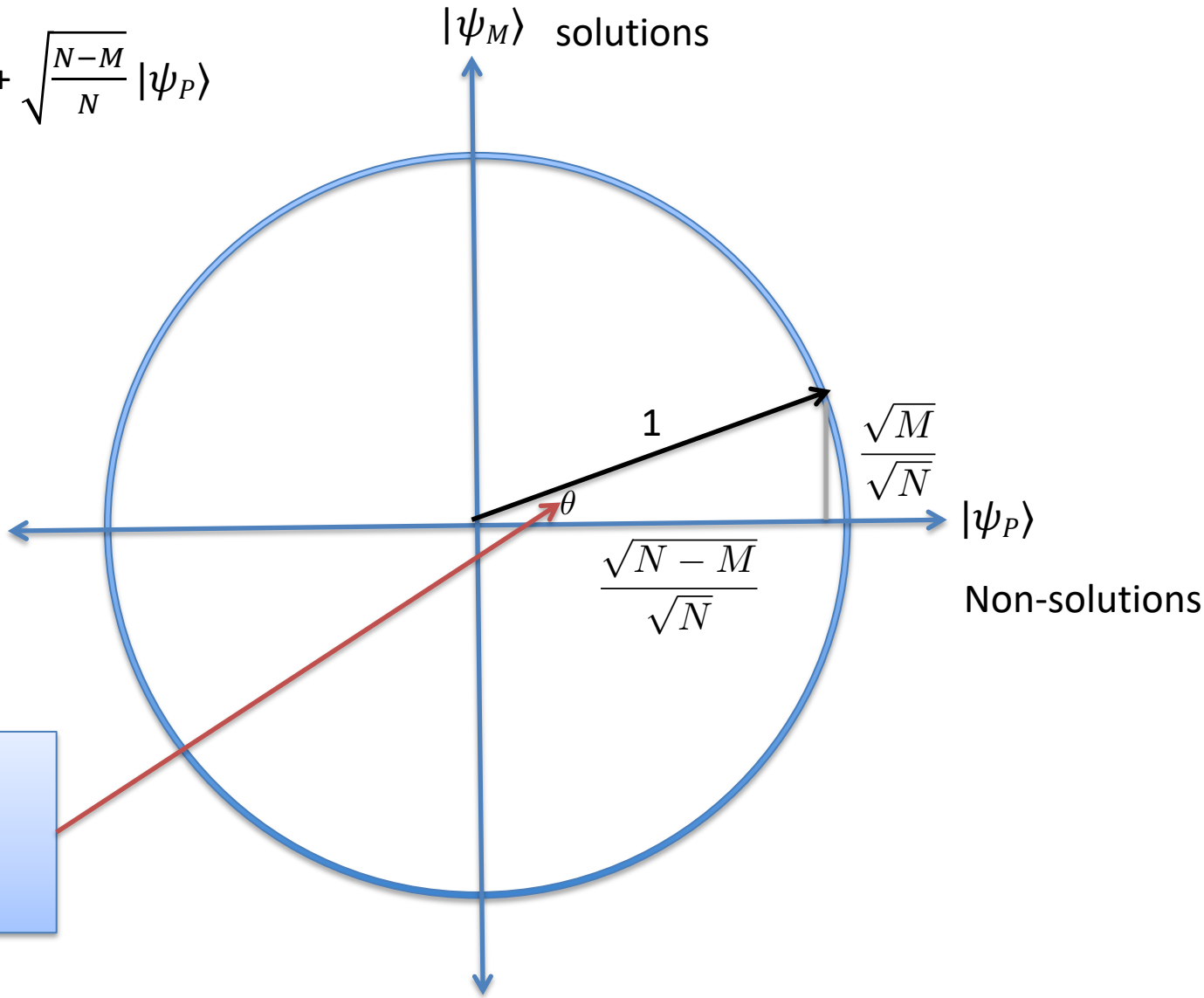
$$k \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}$$

Having multiple solutions is faster than searching for a single marked solution.

Recap: Geometric interpretation

Consider the equal superposition:

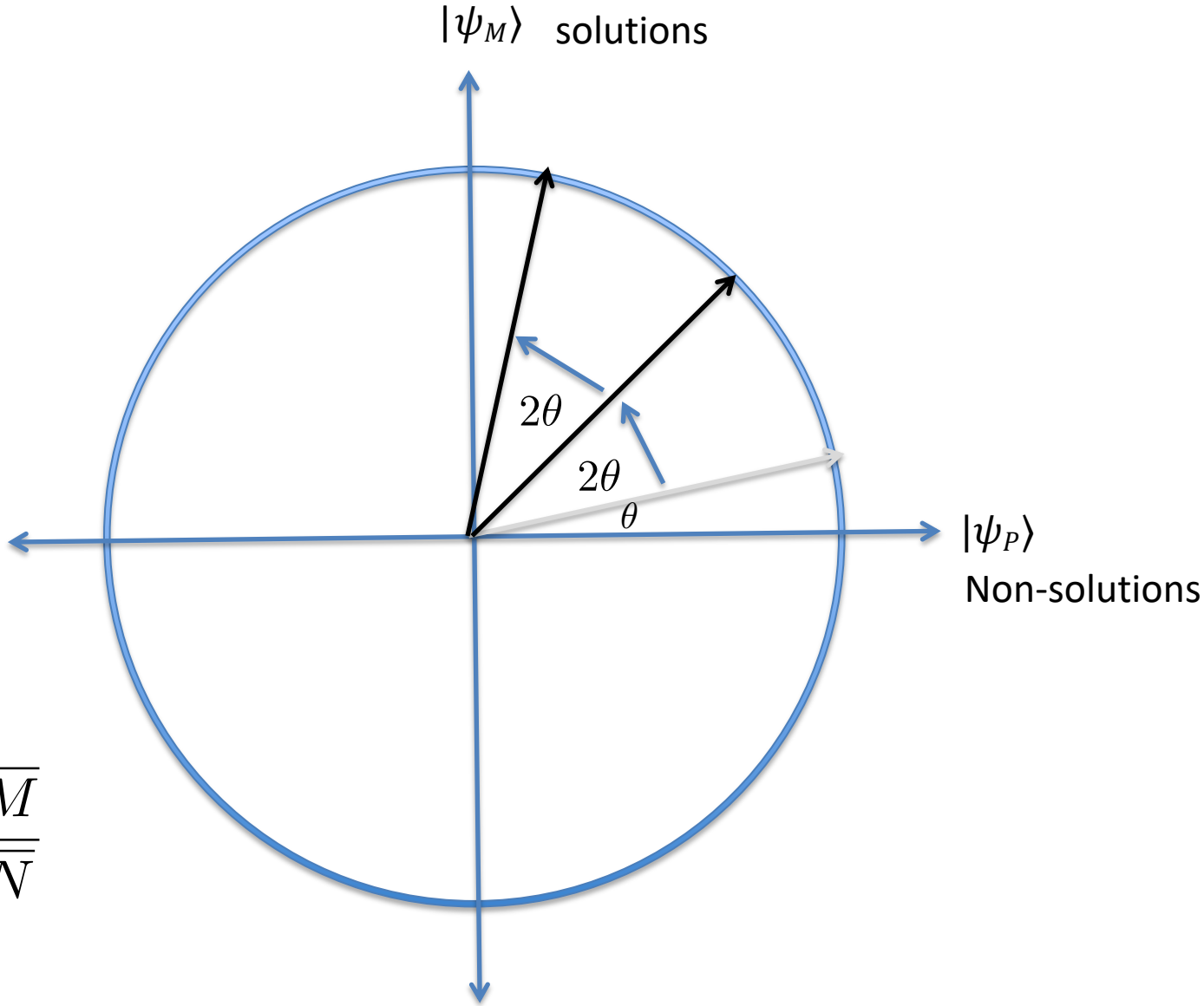
$$|\psi_0\rangle = \sqrt{\frac{M}{N}}\,|\psi_M\rangle + \sqrt{\frac{N-M}{N}}\,|\psi_P\rangle$$



$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

Recap: Geometric interpretation

Product of two reflections is a rotation.



$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

17.1 Unknown number of solutions

MULT20015

Lecture 17

Grover's algorithm – Unknown number of solutions

- The optimal number of Grover iterations depends on the number of solutions M relative to the size of search space N.
  - In general, M is unknown!

- Two approaches to account for unknown number of solutions:

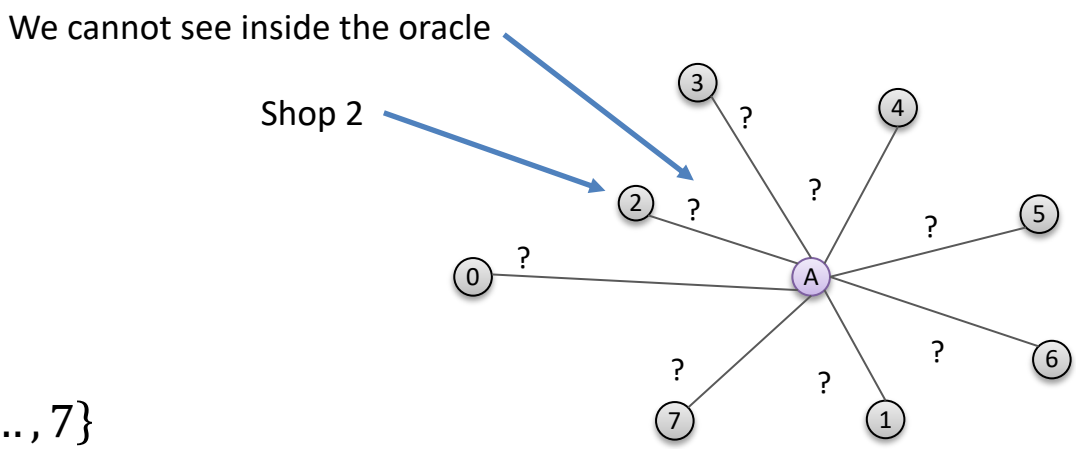
1. Execute Grover's algorithm with number of iterations i selected randomly between 1 and $\frac{\pi}{4}\sqrt{N}$
2. Count the number of solutions using Quantum counting

Note: One of the assumptions of Grover's algorithm was $M < \mathrm{N}$.

- Since M is unknown, it could be that the problem has many solutions.
- To account for this, we can double the search space to $2N$ by one extra "dummy" qubit.

Alice's coffee example

- Example: Alice wants to find a coffee shop that has nice cakes.



- $X = \{0,1, \dots, 7\}$
  - X is not ordered
- $f(x) = \begin{cases} 1, \text{ if solution} \\ 0, \text{otherwise} \end{cases}$
  - In general, the function f does not need to know about the search space.
- We don't know how many solutions are there.
- The function f is a blackbox (similar to Deutsch-Jozsa and Simon's algorithms).

17.2 Quantum counting

MULT20015

Lecture 17

Grover iteration as Matrix operation

- We can write the state after j iterations of Grover as:

$$G^j|\psi_0\rangle = \cos((2j+1)\theta)\,|\psi_P\rangle + \sin((2j+1)\theta)|\psi_M\rangle$$

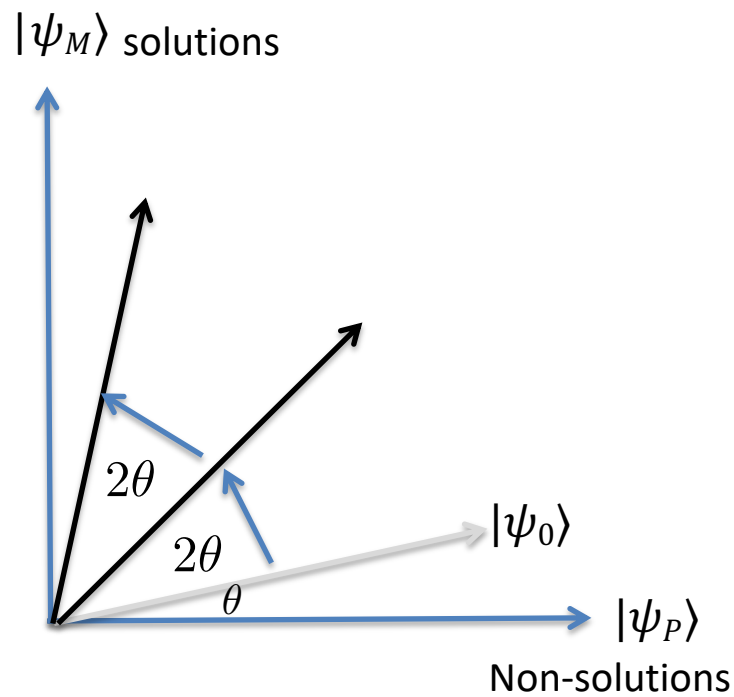
- In the basis $|\psi_M\rangle, |\psi_P\rangle$ Grover's iteration can be written as:

$$G = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix} \text{ where, } \sin\theta = \sqrt{\frac{M}{N}}$$

- Matrix $G = \begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ has

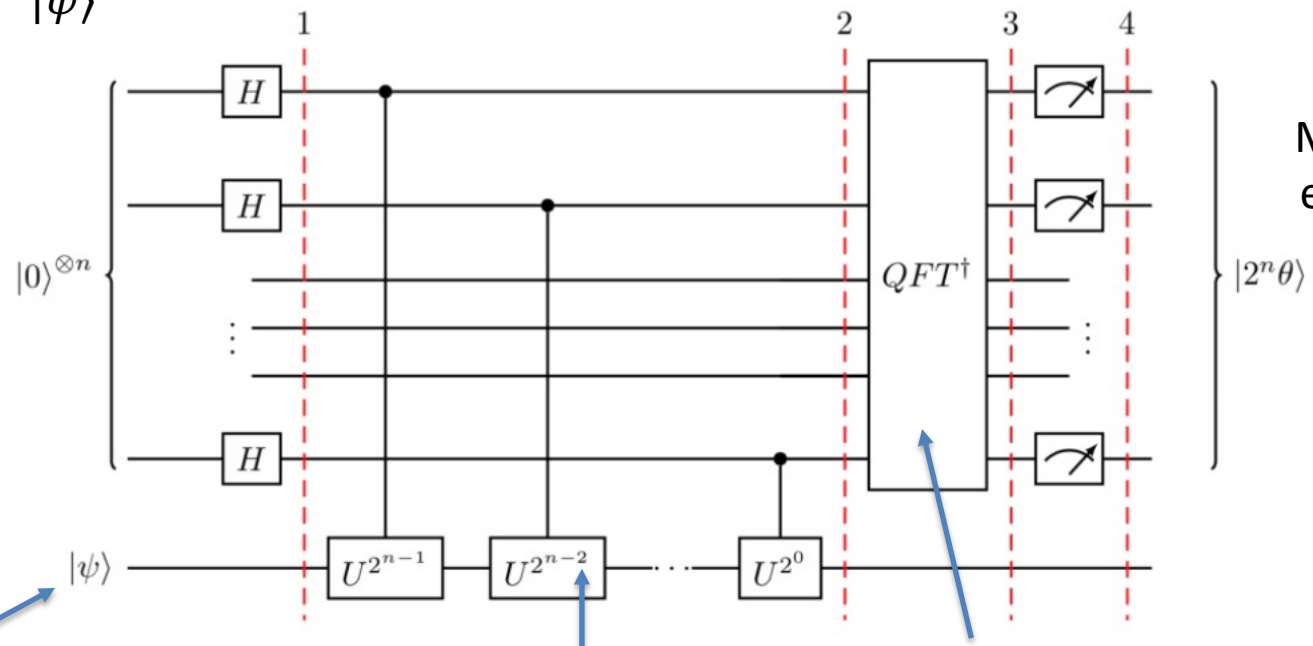
  two eigenvalues $e^{2i\theta}$ and $e^{-2i\theta}$

We can measure θ by Quantum phase estimation.



Quantum Phase Estimation Circuit

$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$



Measure an estimate of $2^n\theta$

Eigenstate

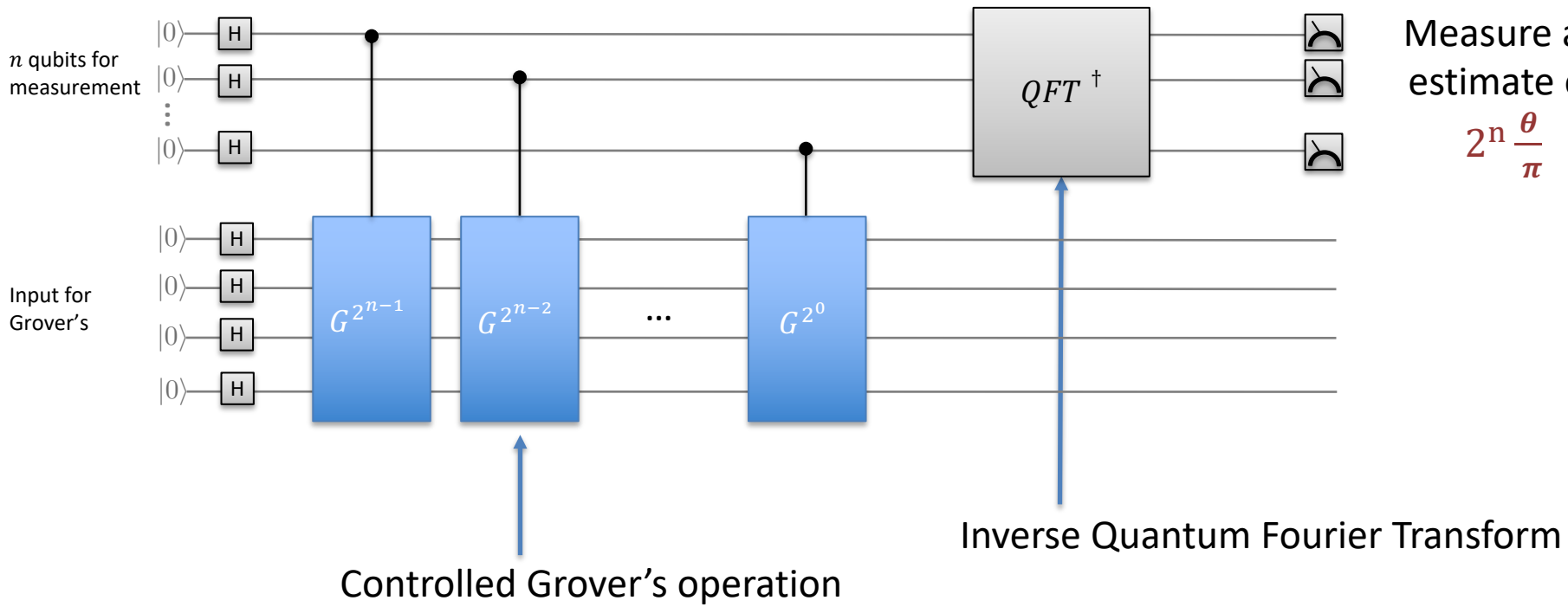
Controlled operation, U

Inverse Quantum Fourier Transform

(1) Prepare equal superposition, x in the upper register, and eigenstate $|\psi\rangle$.
(2) Apply U^x to lower register
(3) Apply (inverse) Quantum Fourier Transform to the upper register
(4) Measure to obtain an estimate equal to $2^n\theta$.

Quantum Counting Circuit

$$U|\psi\rangle = e^{\pm 2i\theta}|\psi\rangle$$



(1) Prepare equal superposition, x in the upper register, and eigenstate $|\psi\rangle$.
(2) Apply G^x to lower register
(3) Apply (inverse) Quantum Fourier Transform to the upper register
(4) Measure to obtain an estimate equal to $\frac{2^n\theta}{\pi}$.

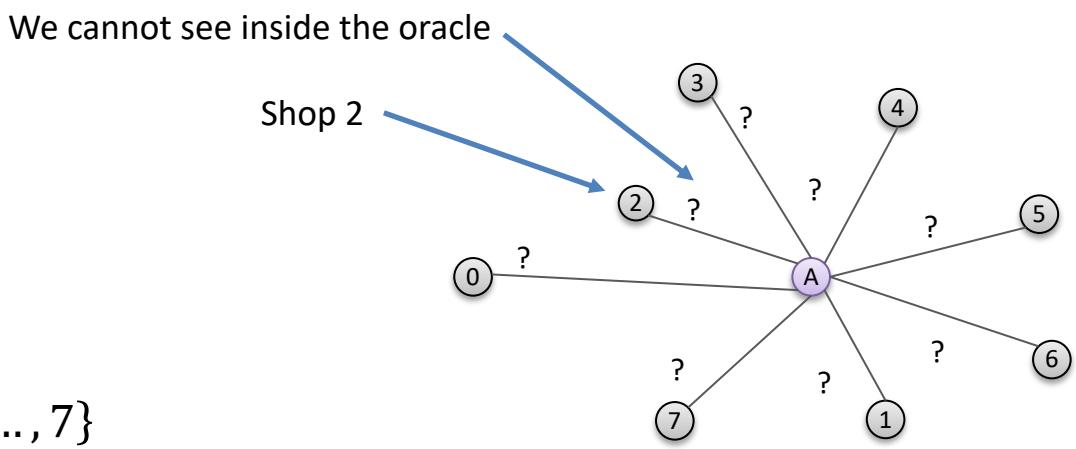
17.2 Quantum counting example

MULT20015

Lecture 17

Alice's coffee example

- Example: Alice wants to find a coffee shop that has nice cakes.



- $X = \{0,1, \dots, 7\}$
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- $f(x) = \begin{cases} 1, \text{ if solution} \\ 0, \text{otherwise} \end{cases}$
  - In general, the function f does not need to know about the search space.
- We don't know how many solutions are there.
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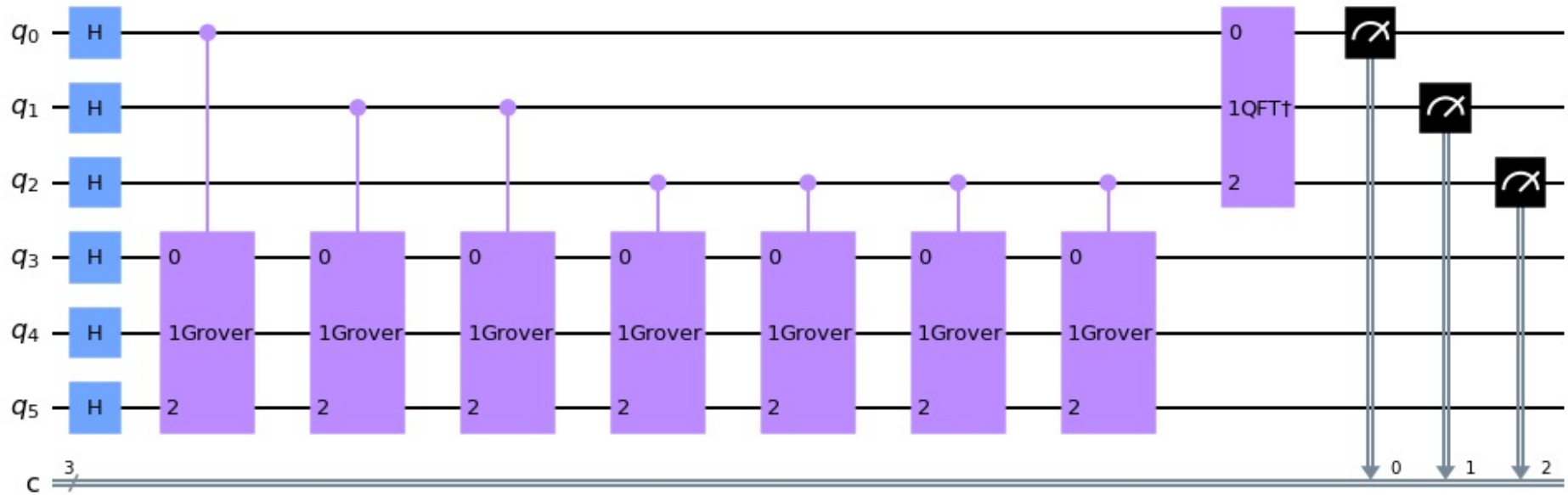
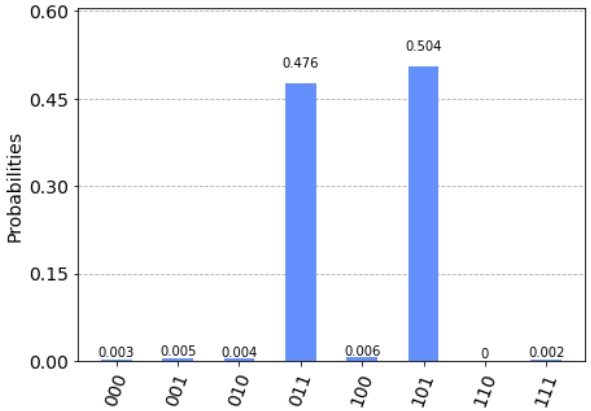
Quantum counting example

- (The figures are from Qiskit) New version for QUI will have this feature.





Measured output = 5 (101 in decimal).

Quantum counting example

- Calculating angle θ by using the measured value:
  - $\frac{2^n\theta}{\pi} = 5$, alternatively,
  - $\theta = 5\pi/2^3$ (we used 3 bits for counting)
  - $\theta = 1.963495$

- The value of θ can respond to either of the eigenvalues! (that is, number of solutions or number of non-solutions).

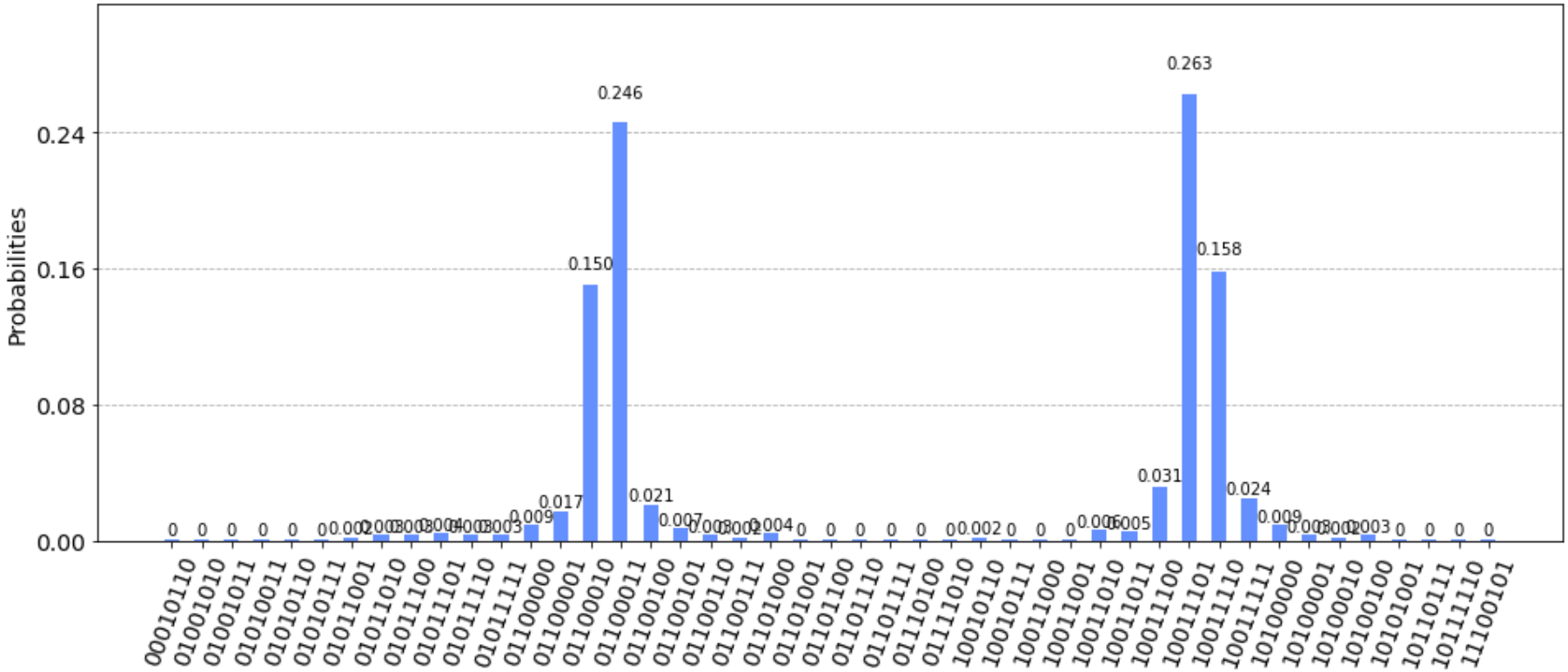
- We know that $\sin\theta = \sqrt{\frac{M}{N}}$ or $\cos\theta = \sqrt{\frac{M}{N}}$
  - $M = 8 * \sin^2(1.963495)$ or $M = 8 * \cos^2(1.963495)$
  - $M = 6.828$ or $M = 1.172$

- Since our assumption was $M << N$, therefore $M = 1.172 \sim 1$

- There is only one coffee shop that serves cake!

Quantum counting example

- Let us increase the number of counting bits to 8



Measured output = 157 (10011101 in decimal).

Quantum counting example

- Calculating θ by using the measured value:
  - $\frac{2^n\theta}{\pi} = 157$, alternatively,
  - $\theta = 157\pi/2^8$ (we used 8 bits for counting)
  - $\theta = 1.926679$

- The value of θ can respond to either of the eigenvalues! (that is, number of solutions or number of non-solutions).

- We know that $\sin\theta = \sqrt{\frac{M}{N}}$ or $\cos\theta = \sqrt{\frac{M}{N}}$
  - $M = 8 * \sin^2(1.926679)$ or $M = 8 * \cos^2(1.926679)$
  - $M = 7.028$ or $M = 0.972$

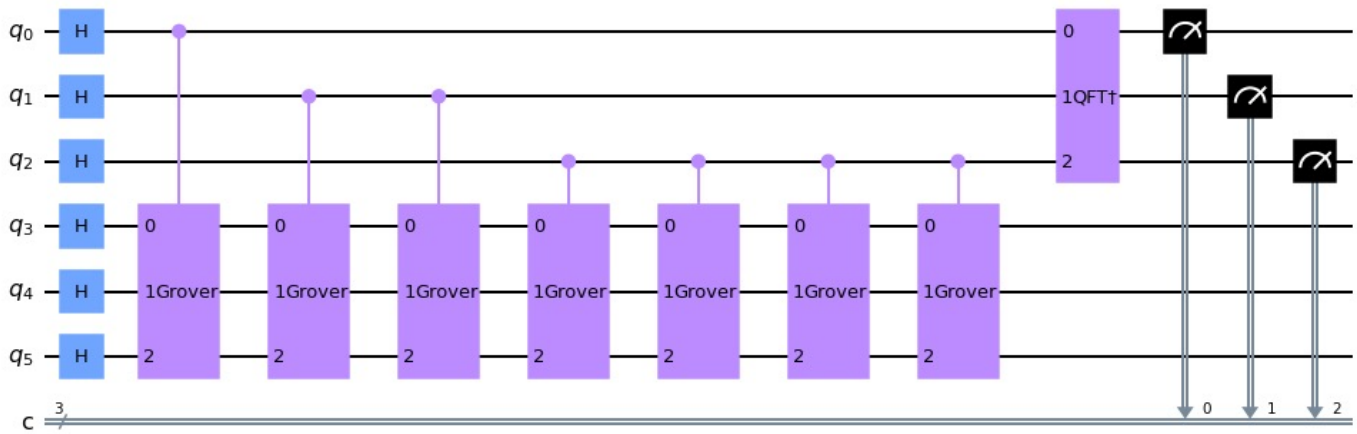
We get a lower error now!

- Since our assumption was $M << N$, therefore $M = 0.972 \sim 1$

- There is only one coffee shop that serves cake!

Grover's with unknown solutions: Example

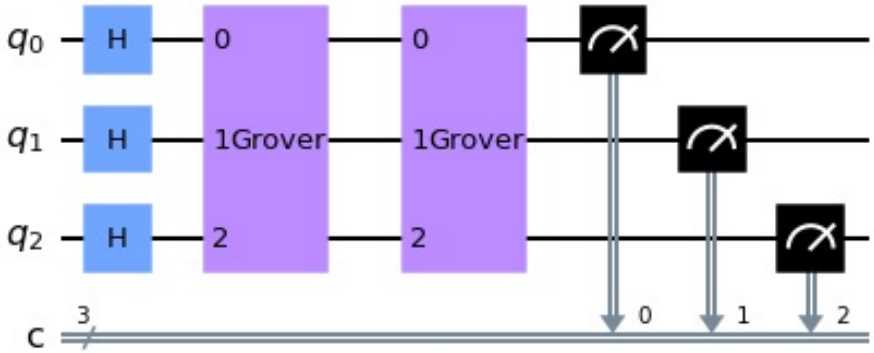
Step 1: Estimate the number of solutions M by quantum counting



$M = 1$

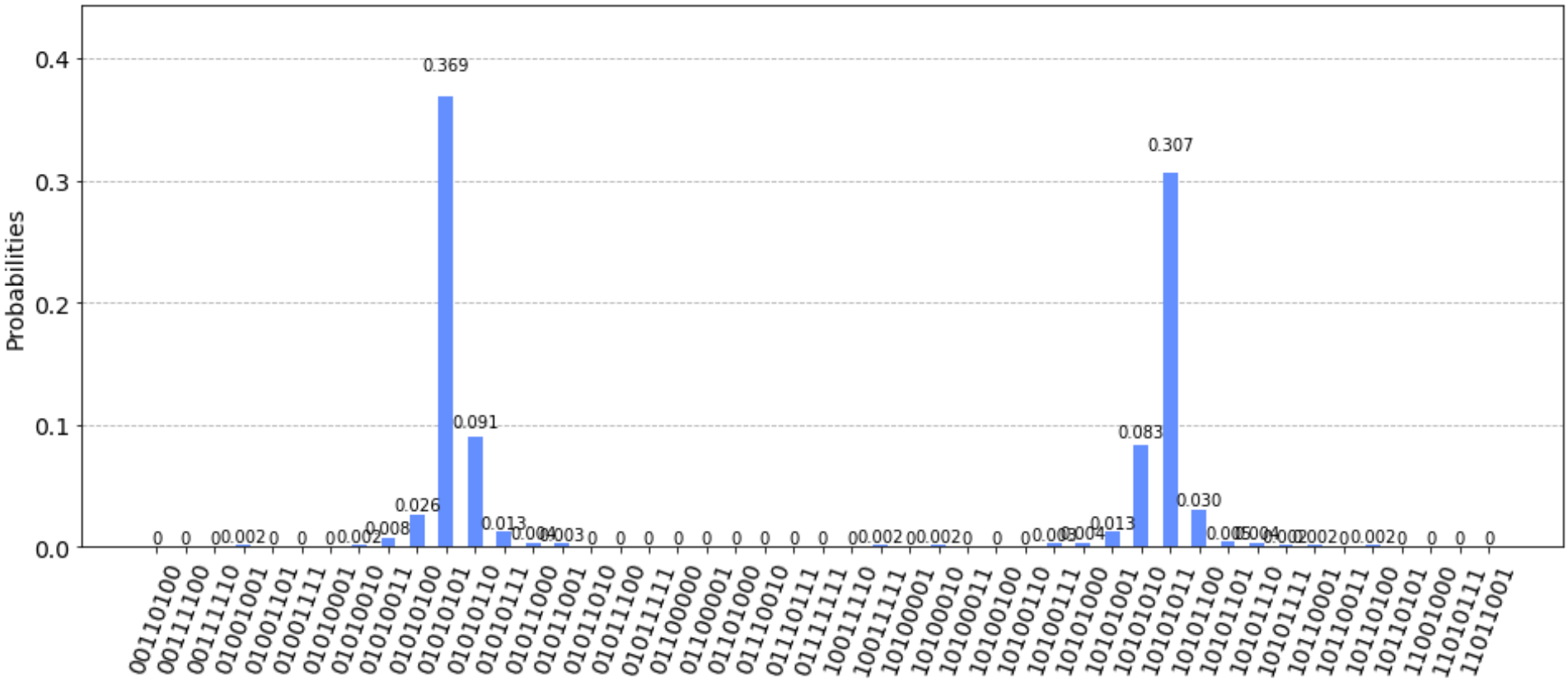
Step 2: Number of Grover iterations $\sin\theta = \sqrt{\frac{M}{N}} = \sqrt{\frac{1}{8}}$, or $\theta = 0.361367$.

$(2k+1)\theta = \pi/2$, or $k = 2$



Quantum counting example: multiple solutions

- Using 8 counting bits



Measured output = 85 (01010101 in decimal).

Quantum counting example

- Calculating θ by using the measured value:
  - $\frac{2^n\theta}{\pi} = 85$, alternatively,
  - $\theta = 85\pi/2^8$ (we used 8 bits for counting)
  - $\theta = 1.0431$

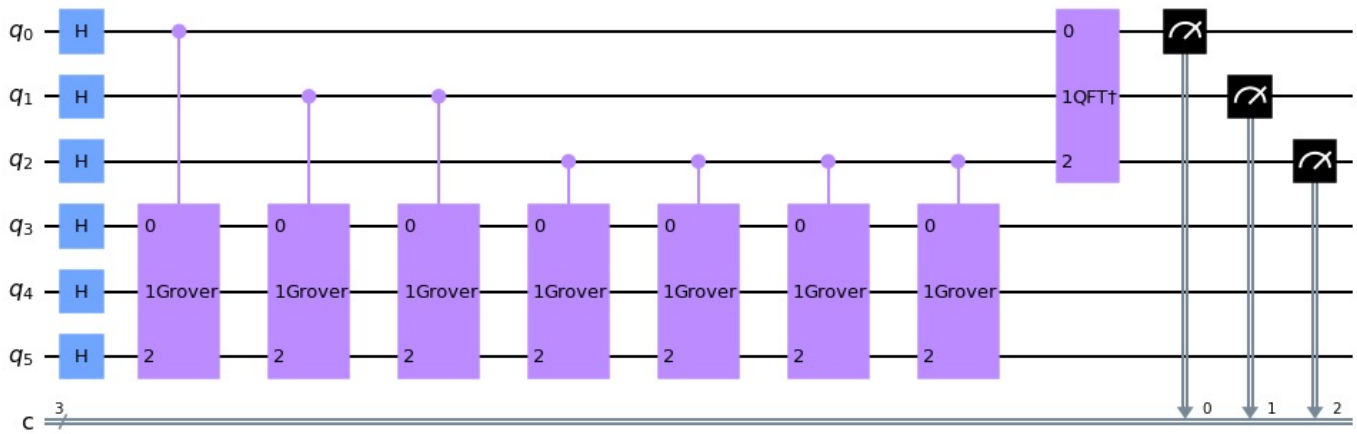
- The value of θ can respond to either of the eigenvalues! (that is, number of solutions or number of non-solutions).

- We know that $\sin\theta = \sqrt{\frac{M}{N}}$ or $\cos\theta = \sqrt{\frac{M}{N}}$
  - $M = 8 * \sin^2(1.0431)$ or $M = 8 * \cos^2(1.0431)$
  - $M = 5.97159$ or $M = 2.02841$

- Since our assumption was $M << N$, therefore $M = 2.02841 \sim 2$

Grover's with unknown solutions: Example

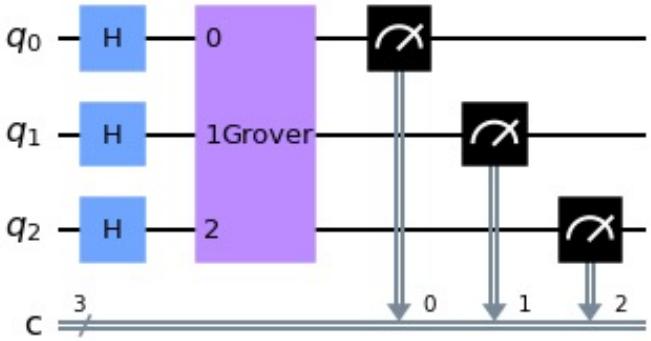
Step 1: Estimate the number of solutions M by quantum counting



$M = 2$

Step 2: Number of Grover iterations $\sin\theta = \sqrt{\frac{M}{N}} = \sqrt{\frac{2}{8}}$, or $\theta = 0.52359$.

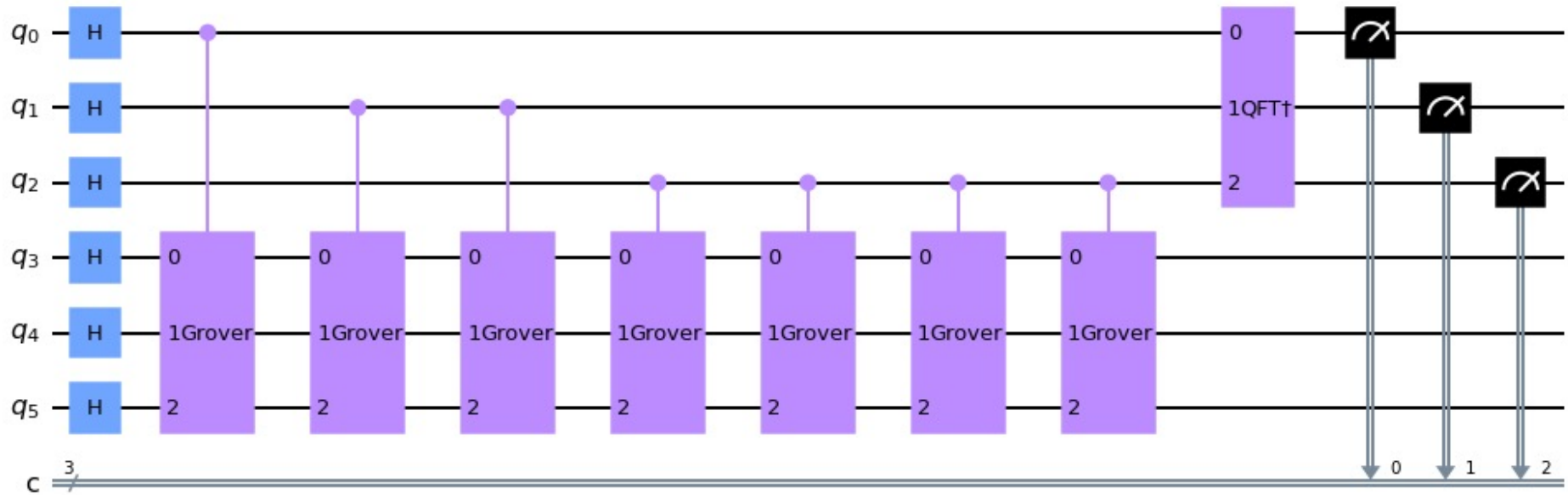
$(2k+1)\theta = \pi/2$, or $k = 1$

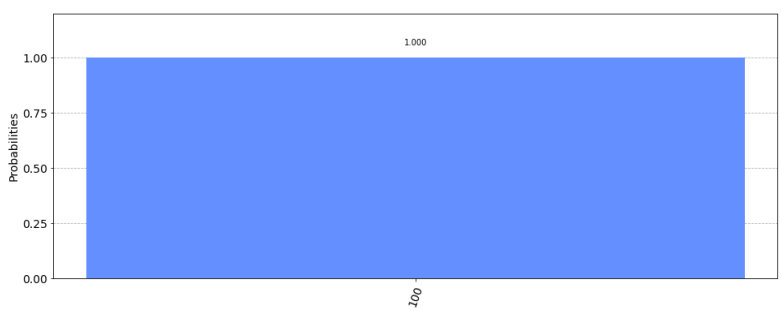


Grover's algorithm and Quantum counting

- Grover's algorithm solves a blackbox problem with only $\sqrt{N}$ calls to an oracle.
- Provides a quadratic speedup over classical methods
  - If there is no structure that classical algorithms can exploit!
- Grover's algorithm is optimal amongst other Quantum computing techniques
  - This is the best we can hope for!
- Grover's algorithm is sometimes referred as a database search algorithm
  - This is misleading as databases often have structure
- Quantum counting has applications beyond computing the optimal number of Grover iterations.
  - Useful for NP problems where one is interested in existence of solutions.
- Phase estimation and amplitude amplification relevant for option pricing
  - Pricing options by simulating possible price trajectories

Quantum counting example: No solution





Measured output = 4 (100 in decimal).

Quantum counting example

- Calculating angle θ by using the measured value:
  - $\frac{2^n\theta}{\pi} = 4$, alternatively,
  - $\theta = 4\pi/2^3$ (we used 3 bits for counting)
  - $\theta = 1.57079$

- The value of θ can respond to either of the eigenvalues! (that is, number of solutions or number of non-solutions).

- We know that $\sin\theta = \sqrt{\frac{M}{N}}$ or $\cos\theta = \sqrt{\frac{M}{N}}$
  - $M = 8 * \sin^2(1.57079)$ or $M = 8 * \cos^2(1.57079)$
  - $M = 8$ or $M = 0$

- Since our assumption was $M << N$, therefore $M = 0$

- There is no solution!

Week 9

Lecture 17

17.1 Grover's algorithm with Unknown number of solutions

17.2 Quantum counting

17.3 Quantum counting example

Lecture 18 Quadratic Unconstrained Binary Optimization (QUBO)

18.1 Optimization Problems

18.2 Number Partitioning

18.3 Graph Partitioning

18.4 Traveling Salesman Problem

Practice class 9

Quantum counting and QUBO

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